CS & AI · EXPLAINED

What is machine learning, simply?

How a computer learns without being told the rules.

THE HOOK

Normally you control a computer by giving it exact rules. Do this, then this.

But how would you write the rules for "is this a cat?" — list every whisker, ear and pose?

Impossible. So we flip the whole approach. 🐱

TODAY'S QUESTION

How can a computer learn something — without anyone writing the rules?

IDEA 1 / 4 — EXAMPLES, NOT RULES

Stop writing rules. Show examples.

```
# traditional programming
Rules    →  computer  →  Answers

# machine learning
Examples →  computer  →  Rules
```

- Old way: **you** write the rules, the computer just follows.
- ML way: you show lots of examples...
- ...and the computer figures out the rules **by itself**.

Show it thousands of labelled photos.

🐱 cat | 🐶 dog | 🐱 cat | 🐱 cat | 🐶 dog | … ×1000s

- It gradually finds the patterns on its own — shapes, ears, textures.
- Just like a small child learns ”cat” from **seeing many cats** — not from a definition.
- This phase is called **training**.

IDEA 3 / 4 — PREDICTION ON NEW DATA

Now it can handle a brand-new photo.

🖼️ never seen before → trained model → 🐱 ”cat” (92%)

- It looks at a photo it has **never seen** and predicts ”cat”.
- Based on the patterns it learned during training.
- That’s the whole point: it **generalises** to new examples.

It’s making an **informed guess** — not ”knowing”.

IDEA 4 / 4 — ONLY AS GOOD AS THE DATA

Garbage in, garbage out.

- A model is only as good as its examples.
- Show it **biased** or **wrong** data…
- …and it learns biased or wrong patterns.

WHY IT MATTERS

This is why **data quality** matters so much.

WHY IT MATTERS

ML is already everywhere.

- 🔓 Face unlock on your phone.
- 📧 Spam filters in your inbox.
- ▶️ YouTube & Instagram recommendations.
- 💳 Fraud detection on your bank card.

CALLBACK — EP01

Chatbots like ChatGPT? That’s ML too.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

✗ MYTH

"The machine **understands** and **thinks** like us."

✓ REALITY

It found **statistical patterns** in examples. No understanding — just very powerful `pattern-matching` and guessing.

RECAP — 2 THINGS TO REMEMBER

1. ML = learning rules from examples, instead of being told the rules.
2. It predicts on new data — and it’s only as good as the examples you gave it.

Learn from examples, guess on new ones. 🧠

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Next question: **How does your phone know your location (GPS)?** 📍